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Modeling magnetic braking in the Pleiades cluster

Introduction

The goal of this project is to model the rotational evolution of stars over time. By combining known relationships (such as mass–radius scaling and color-based mass estimates) with physical laws like angular momentum conservation and magnetic braking, we can simulate how a star's rotation rate changes as it ages. Numerical methods were used to solve the governing differential equations, with the aim of reproducing realistic spin evolution and exploring how stellar properties influence rotation behavior.

Theory

As stars lose mass, they must rotate faster in order to conserve angular momentum, similar to a figure skater pulling their arms closer to their body, but this is not the case, signaling that there must be a counteracting force (Wikimedia, 2026). The model accounts for these two competing effects. Magnetic braking is a phenomenon in which celestial bodies lose angular momentum over time due to plasma ejections and interactions within the body's magnetic field. Mathematically, this can be represented as a derivation of Newton's Second Law, with torque being the rotational equivalent of force. The magnetic field acts as a lever arm while the torque results from the stellar wind. Stellar contraction is the term that describes the increase of a star's rotational velocity as its moment of inertia decreases.

0.1 Color and Mass Relation

The stellar mass is estimated from the color index $V - K$:

$$M = 1.2 - 0.15\,[(V - K) - 1.0] \tag{1}$$

The constraint for this relation is:

$$1.0 \le (V - K) \le 4.0 \tag{2}$$

Where: - M: stellar mass (in solar masses $M_\odot$) - $V - K$: color index (visual − infrared magnitude) (Ardestani, 2017)

0.2 Mass and Radius Relation (ZAMS)

$$R_{\text{ZAMS}} = M^{0.8} \tag{3}$$

Where: - R_{ZAMS}: zero-age main sequence radius (in solar radii)

0.3 Time-Dependent Radius Evolution

Young stars start larger and contract over time:

$$R_{\text{early}} = 1.5\, R_{\text{ZAMS}} \tag{4}$$

$$R(t) = R_{\text{ZAMS}} + (R_{\text{early}} - R_{\text{ZAMS}})\, e^{-t/\tau} \tag{5}$$

Where: - t: time (Myr) - $\tau = 30$ Myr (contraction timescale)

0.4 Moment of Inertia

$$I(t) = k^2 M R(t)^2 \tag{6}$$

Where: - $I(t)$: stellar moment of inertia - $k^2 = 0.2$: structure constant

0.5 Angular Momentum

$$L = I\omega \tag{7}$$

Where: - L: angular momentum - ω: angular velocity

0.6 Torque and Spin Evolution

Magnetic braking law:

$$\frac{d\omega}{dt} = -\frac{K}{I(t)}\omega^n \tag{8}$$

Where: - K: torque constant - n: braking exponent

0.7 Full Time-Dependent Spin Evolution Equation

Including stellar contraction:

$$\frac{d\omega}{dt} = -\frac{K}{I(t)}\omega^n + \omega\frac{1}{I(t)}\frac{dI}{dt} \tag{9}$$

The first term represents magnetic braking (spin-down), and the second term represents the contraction effect (spin-up if $dI/dt < 0$).

0.8 Initial Condition (IVP)

$$\omega(0) = \omega_0 \tag{10}$$

0.9 Angular Velocity from Period

$$\omega_0 = \frac{2\pi}{P_0} \tag{11}$$

Where: - P_0: initial rotation period (days)

0.10 Period Evolution

$$P(t) = \frac{2\pi}{\omega(t)} \tag{12}$$

0.11 Initial Period Distribution

$$P_0 \sim \text{LogNormal}(\mu, \sigma) \tag{13}$$

With parameters:

$$mu = -0.3, sigma = 1.05$$

1 Results and Calculations

The two Monte Carlo graphs seen in Figures 1 and 2 are used to show the range of the lognormal function's variance. The function in Figure 1 has a greater peak than that of the observed K2 data, while the second graph has a lesser peak than the observed K2 data, but a smoother decay when compared to the juxtaposing graph (Rebull, 2016). In both cases, the lognormal function serves as a useful barometer for the shorter rotational period distributions and may potentially be used to model the period distribution for the younger members of other clusters. The right tail of the observed data cannot be accurately represented with only one lognormal function; another distribution function may be needed to more accurately represent the observed period distributions.

The spin evolution graph is derived from an initial value problem that incorporates the time-dependent spin evolution equation. Using this differential equation and parameters matching the age of the Pleiades cluster, the resulting graph in Figure 3 was produced, tracking the angular velocity over the given star's lifetime. This data allows for a comparison with a star reverse-evolved from its current age to the birth of the Pleiades cluster. This comparison not only serves as a simple sanity check but also allows for new insight into the cluster and its formation. This process allows researchers to infer earlier rotational periods, and from this knowledge, it can be estimated how long a given star has been spinning down, which provides the framework for estimates of a star's age. The initial rotational periods can clarify the conditions during the star's formation and the initial conditions of surrounding stars within the cluster. The graph in Figure 4 contains a different star than the graph in Figure 3, as seen by the differing EPIC ID. As this graph contains period data while the other contains velocity data, a quick calculation confirms that the two stars have differing rotational velocities and periods. This sanity check allows for a confirmation that spin evolution over time acts as an exponentially decaying function.

One hypothesis that arose before work began was that the behavior of the spin evolution graph would follow a linearly decreasing trend, represented algebraically by a power law with a negative exponent, likely -0.5, meaning the stars' rotation speed is inversely proportional to the square root of their age (Sayeed 2022). This behavior is seen when simulations are conducted with a time-independent moment of inertia, which logistically is very facile, but this method is physically incorrect over the given timeframe. It must be noted that the graph's trend represents a near constant over 100 million years, with a decrease of only 0.14 radians per day. However, this timeframe is just a small fraction of the median star's lifespan.

The radial contraction over time plot does not serve as a model of stars specific to the Pleiades cluster, but rather represents a generalized range of stellar masses of low-mass pre-main-sequence stars. The simulation uses the 120 Myr age of the Pleiades cluster as a reference, but other masses and time frames would also be valid. In Figure 7, the initial loss of rotational velocity is proportional to the size of the star's radius, with larger masses losing radius at a greater rate than the smaller PMS stars. This is accurate physically, as the larger radius correlates with a larger surface area for magnetic stellar winds to operate.

1.1 Monte Carlo Simulations

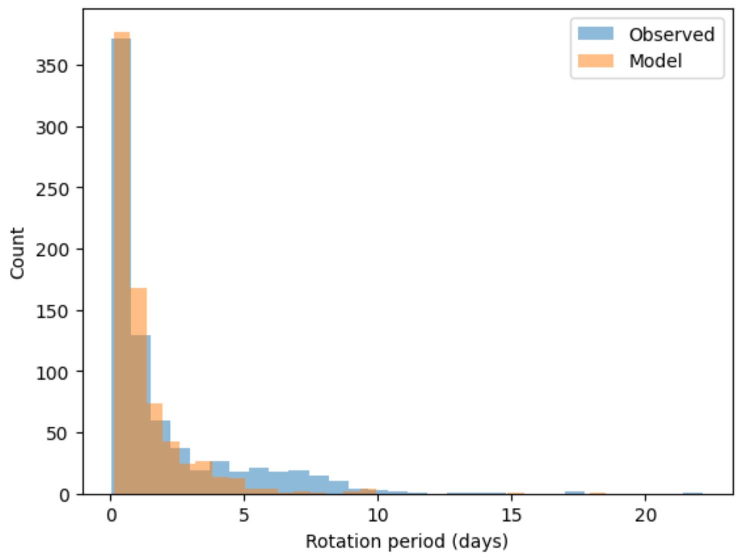

Figure 1: Monte Carlo simulation showing the initial rotational period distribution of stars in a trial.

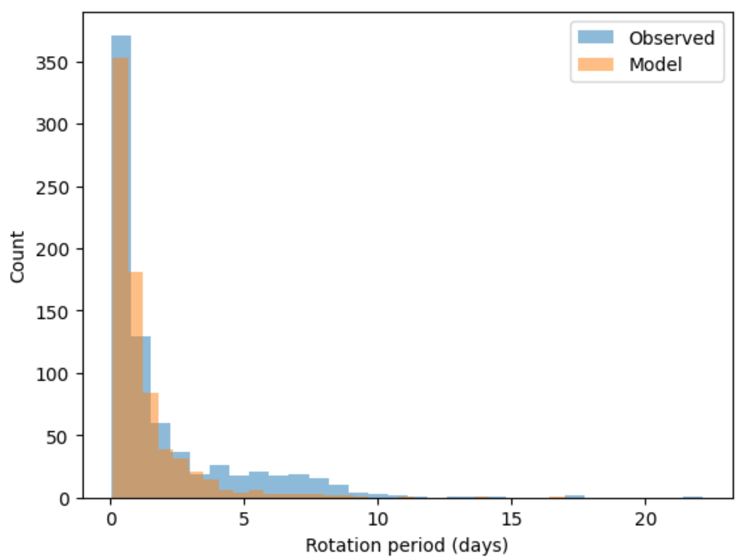

Figure 2: Results of a second Monte Carlo simulation showing the initial rotational period distribution of stars in a trial.

1.2 Spin Evolution

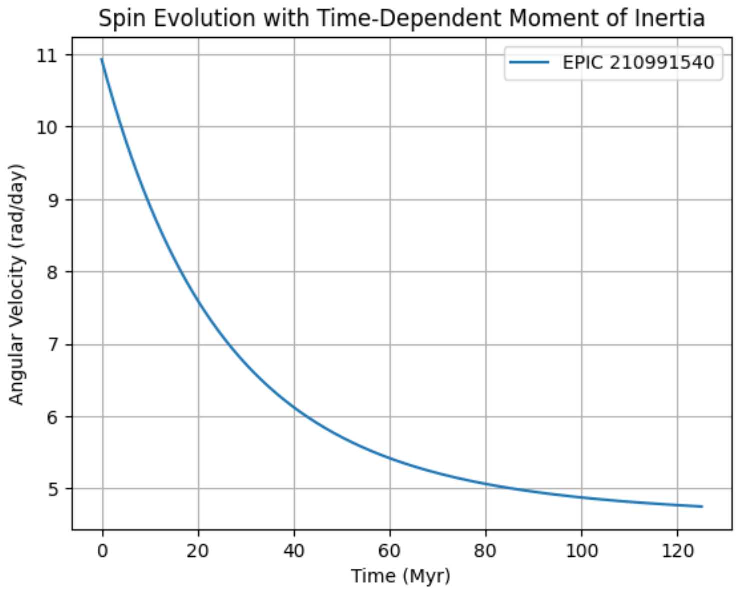

Figure 3: Spin evolution of stars over time under magnetic braking.

1.3 Reverse Spin Evolution of a Star

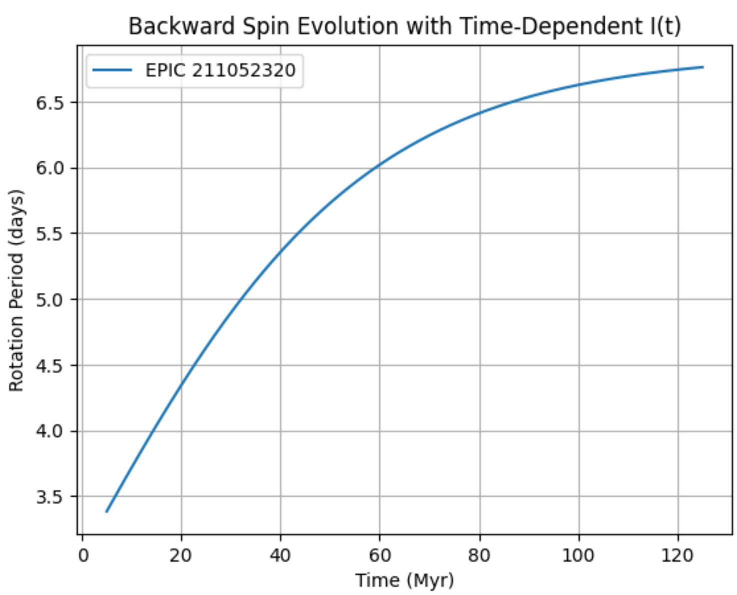

Figure 4: Reverse evolution of a stellar rotation period back toward earlier ages.

1.4 Results with Constant Moment of Inertia

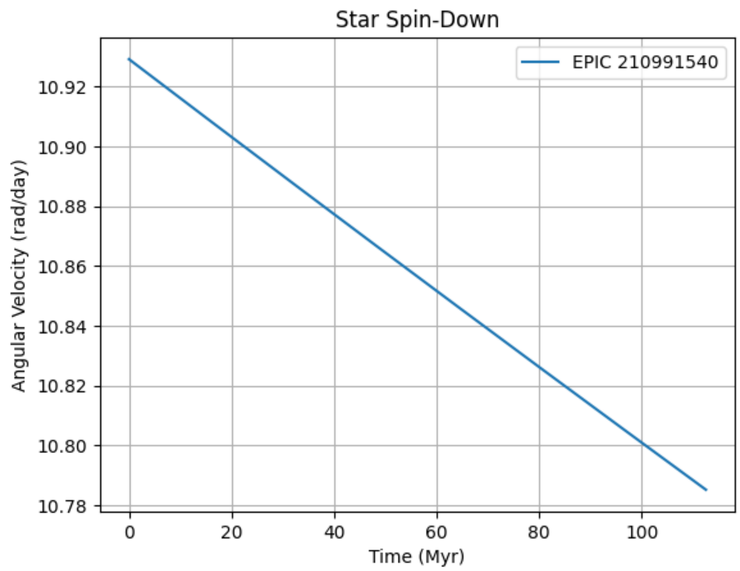

Figure 5: Spin evolution assuming a constant stellar moment of inertia.

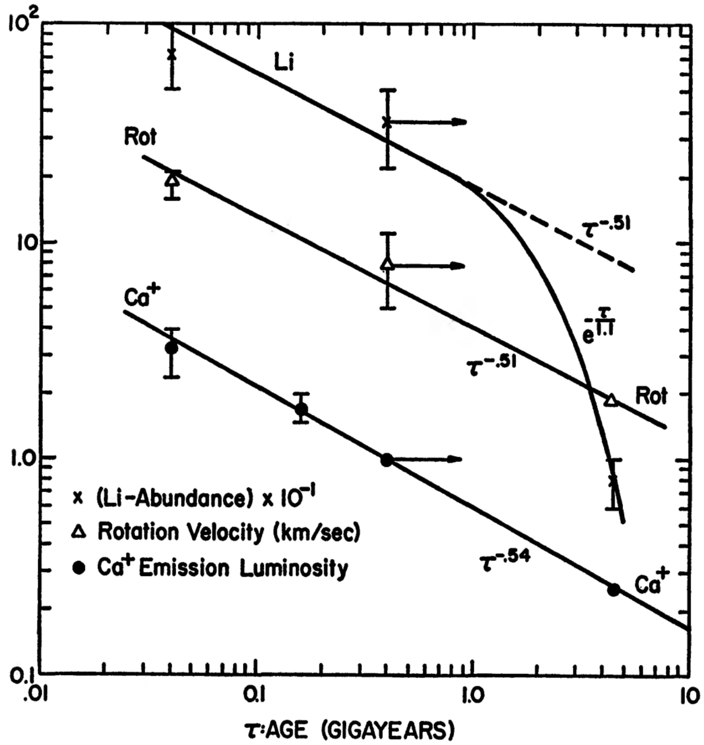

Figure 6: Rotational velocity over time (Skumanich, 1972).

1.5 Radial Contraction Over Time

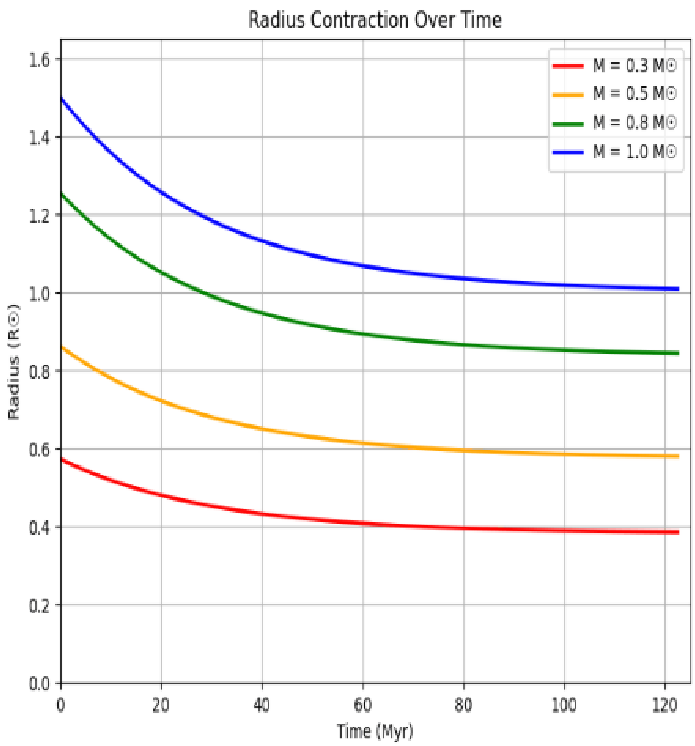

Figure 7: Angular velocity as a function of stellar age.

Discussion

The analysis of the K2 datasets shows that there are multiple rotational sequences, indicating that a single lognormal distribution is insufficient to fully capture the structure of the data. This can be seen particularly via the mismatch between the primary peak and the extended right-side tail. Another function may need to be added to more accurately represent the right side of the data. The current function is useful for accurately representing the skew for the younger portion of the cluster, and may possibly be used as an assumption of this stellar age range for other clusters. Using a Skumanich-type magnetic braking model, stellar spin evolution was shown to follow a consistent exponential decay trend across multiple stars, and reverse-time simulations from any maximum time to t=0 confirm the validity of these results. This knowledge allows us to infer earlier rotational states of stars from their present-day properties. Comparative simulations across different stellar masses show that angular velocity initially decays at similar rates, but lower-mass stars experience more significant long-term spin-down. Additionally, modeling radius evolution demonstrates that higher-mass stars undergo faster early-stage contraction, while stars of all masses eventually converge toward similar, stable contraction rates. Overall, the results highlight the importance of incorporating both magnetic braking and time-dependent structural evolution to accurately model stellar rotation histories.

Conclusion

This project demonstrates that stellar rotational evolution is driven by magnetic braking and radial contraction. While a single lognormal distribution doesn't fully represent the entirety of the observed K2 rotational sequences, the spin-down model successfully reproduces key trends in stellar rotation over time. By applying a Skumanich-type braking law with time-dependent moment of inertia, we are able to both model present-day rotation rates and infer earlier stellar states. The results show that stellar mass plays a significant role, influencing both the rate of angular momentum loss and the timescale of contraction. Overall, incorporating both physical braking mechanisms and stellar evolution effects provides a more realistic and insightful description of stellar spin behavior.

References

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